

Project Delay Tracking System – Functional Requirements

1. Objective

Develop a centralized Project Delay Tracking System to monitor construction and renovation projects across hospitals, identify delay causes, track accountability, estimate financial impact, and provide early warning alerts before delays become critical.

MODULE 1 – PROJECT DELAY TRACKING

2. Project Master

Fields

- Project ID
- Hospital Name
- Project Type
- Green Field
- Brown Field
- Renovation
- Area / Facility
- Project SPOC
- Planned Start Date
- Planned Completion Date
- Revised Completion Date
- Project Status
- Planned
- In Progress
- Delayed
- Completed
- On Hold

Functions

- Create Project
 - Edit Project
 - View Project
 - Close Project
 - Upload project documents
 - Upload before/after photographs
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3. Delay Register

Each project can have multiple delay entries.

Delay Fields

Delay Details

- Delay ID (Auto Generated)
- Project ID
- Delay Category
- Primary Delay Driver
- Specific Event Description
- Impacted Task
- Root Cause
- Responsibility

Delay Categories

- Regulatory Approvals
- Procurement & Supply Chain
- Design & Scope Changes
- Medical Equipment Installation
- Specialized Labor
- Logistical Challenges
- Site Condition Surprises
- Operational Readiness
- Infection Control Compliance
- MEP Delays

Responsibility

- Contractor
- Architect
- Consultant
- Hospital Administration
- Vendor
- Procurement Team
- Operations Team
- Other

Timeline

- Delay Start Date
- Delay End Date
- Total Delay Days (Auto Calculated)

Formula:

Delay Days = End Date - Start Date

4. Delay Severity Classification

System should automatically calculate severity.

Rules

Minor

- 1 to 7 days

Moderate

- 8 to 30 days

Critical

- More than 30 days

Showstopper

- Licensing/Openings affected

Auto Calculation

Severity should update automatically when delay days change.

5. Mitigation Tracking

Fields

- Mitigation Action
- Owner
- Target Resolution Date
- Current Status
- Resolution Remarks

Status Values

- Open
 - In Progress
 - Escalated
 - Closed
-

6. Financial Impact Tracking

Direct Cost

Capture:

- Labor Overrun
- Material Cost Overrun
- Contractor Claims
- Equipment Storage Charges

Opportunity Cost

Capture:

- Delayed Admissions
- Delayed Surgeries
- Delayed Revenue
- Lost Operational Days

Calculations

Total Project Delay Cost = Direct Cost + Opportunity Cost

7. Attachments

Support multiple uploads:

- Photos
 - Drawings
 - NOCs
 - Approval Letters
 - Vendor Communications
 - Change Orders
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MODULE 2 – EARLY WARNING SYSTEM (EWS)

8. Delay Prediction Engine

Purpose: Identify likely delays before they occur.

Rule

If:

Task Completion < 50%

AND

Consumed Duration > 80%

Then:

Generate Potential Delay Alert

Example:

Allocated Duration = 10 days

Elapsed = 8 days

Progress = 45%

Result:

Potential Delay Alert

9. Alert Levels

Green

On Track

Amber

Potential Delay

Red

Critical Delay

Black

Showstopper

10. Escalation Matrix

Level 1

Project SPOC

Level 2

Department Head

Level 3

Project Steering Committee

Level 4

Management

Escalation should trigger automatically based on delay severity.

MODULE 3 – RENOVATION PROJECT MONITORING

11. Renovation Project Master

Fields:

- Project ID
 - Project Name
 - Project Scope
 - Location
 - Zone / Department Impacted
 - Renovation Type
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12. Task Tracking

Each renovation project contains multiple tasks.

Fields

- Task Category
- Task Description
- Priority
- Planned Start Date

- Planned End Date
- Actual Start Date
- Actual End Date

Priority Values

- High
 - Medium
 - Low
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13. Operational Impact Assessment

Shutdown Required

Values:

- Yes
- No

Patient / Service Disruption Score

Scale:

1 – 10

Temporary Relocation Needed

Values:

- Yes
- No

Infection Control Clearance

Values:

- Pending
 - Approved
 - Rejected
-

14. Dependency Management

Fields:

- Dependency Task

- Dependency Status

System should highlight tasks blocked by pending dependencies.

15. Risk Scoring

Calculate risk using:

- Delay Days
- Disruption Score
- Approval Delays
- Material Delays
- Dependency Delays

Output:

- Low Risk
 - Medium Risk
 - High Risk
 - Critical Risk
-

16. Procurement Tracking

Fields:

- Vendor / Contractor
- Procurement Status
- Material Delay Days

Procurement Status:

- Pending
 - Ordered
 - In Transit
 - Delivered
 - Installed
-

17. Approval Tracking

Fields:

- Approval Status
- Approval Pending Days

Statuses:

- Pending
- Approved
- Rejected

Pending Days should be auto-calculated.

18. Change Order Tracking

Fields:

- Number of Change Orders
 - Change Description
 - Approval Status
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19. Cost Tracking

Fields:

- Budgeted Cost
- Actual Cost
- Cost Overrun %

Formula:

Cost Overrun % = ((Actual Cost - Budgeted Cost) / Budgeted Cost) * 100

20. Daily Delay Log

Capture:

- Date
- Delay Reason
- Entered By
- Corrective Action

Maintain complete delay history.

MODULE 4 – DASHBOARDS & REPORTS

21. Executive Dashboard

KPIs

- Total Projects
 - Active Projects
 - Delayed Projects
 - Completed Projects
 - Average Delay Days
 - Delay Cost
 - High Risk Projects
 - Pending Approvals
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22. Delay Analytics

Reports:

- Delay Category Analysis
 - Delay Trend by Month
 - Delay by Hospital
 - Delay by Contractor
 - Delay by Responsibility
 - Delay Cost Analysis
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23. Renovation Dashboard

KPIs

- Tasks Completed
 - Tasks Delayed
 - High Risk Tasks
 - Pending Approvals
 - Material Delays
 - Infection Clearance Pending
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24. Notifications

System should notify users for:

- New Delay Entry
- Delay Escalation
- Approval Pending > X Days
- Critical Risk Detection
- Project Completion Date Missed

Notifications through:

- Email
 - In-App Alerts
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25. Audit Trail

Track:

- Created By
- Created Date
- Modified By
- Modified Date

Maintain full history of changes.